

USER MANUAL

Compounding Calculator

How to plan a core + satellite portfolio, set a goal, record each year's results and see what starting early is worth.

Version **1.11** Released **24 Oct 2026** Developed by **Sathiya Sekhar E**

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01

About the calculator

The Compounding Calculator projects how a long-term equity portfolio grows when it is split into a steady **core** and a faster-growing **satellite**, and how much of the final money comes from compounding rather than from what you put in.

It answers four questions, one on each page:

Page	The question it answers
Core engine	If I invest this lump sum and this monthly SIP, split this way, what will I have in *n* years?
Goal & schedule	When does my plan reach my goal, what SIP or return would get there on time, and what does every year look like?
Annual review	I have real results for the years so far. Am I still on track, and what must change from here?
Youngster lab	How much is starting at 21 worth compared with starting at 24, 27 or 30?

The core + satellite idea

Most of the money (by default 70%) goes into a **core** of steady compounders expected to earn a moderate return (15% a year by default). The rest (30%) goes into a **satellite** of higher-growth ideas expected to earn more (25%). The calculator grows the two as separate pots and never rebalances them, so over long periods the satellite usually ends up as most of the wealth.

PROJECTIONS, NOT PROMISES

Every result assumes the same return every year. Real markets are uneven, and a bad year early on changes the outcome. Use the Annual review page each year to replace assumptions with what actually happened.

Two editions

	Windows app	Web page
How you open it	Start menu or desktop shortcut	A link in any browser
Calculations	Identical in both	
Users and records	This browser profile, plus an optional database file	That browser only
PDF report, CSV, export of users	Yes	Not available
Deleting with confirmation	A confirmation box	Click the same button twice within 6 seconds

02

Installing and opening

Windows app

1. Run **CompoundingCalculator-Setup.exe**. No administrator rights are needed; it installs only for your Windows user.
2. Click **Yes** to install (or to update an older version). Your users and records are kept during an update.
3. Open **Compounding Calculator** from the Start menu or the desktop. It opens in its own window using Microsoft Edge (or Chrome if Edge is missing).

Program folder

`%LOCALAPPDATA%\Programs\Compounding Calculator`

Data folder

`C:\Users\<you>\Compounding Calculator Data`, created for an optional database file (see chapter 8)

Uninstall

Windows **Settings → Apps → Installed apps → Compounding Calculator → Uninstall**, or run `Uninstall.exe` in the program folder. The data folder is left in place.

Web page

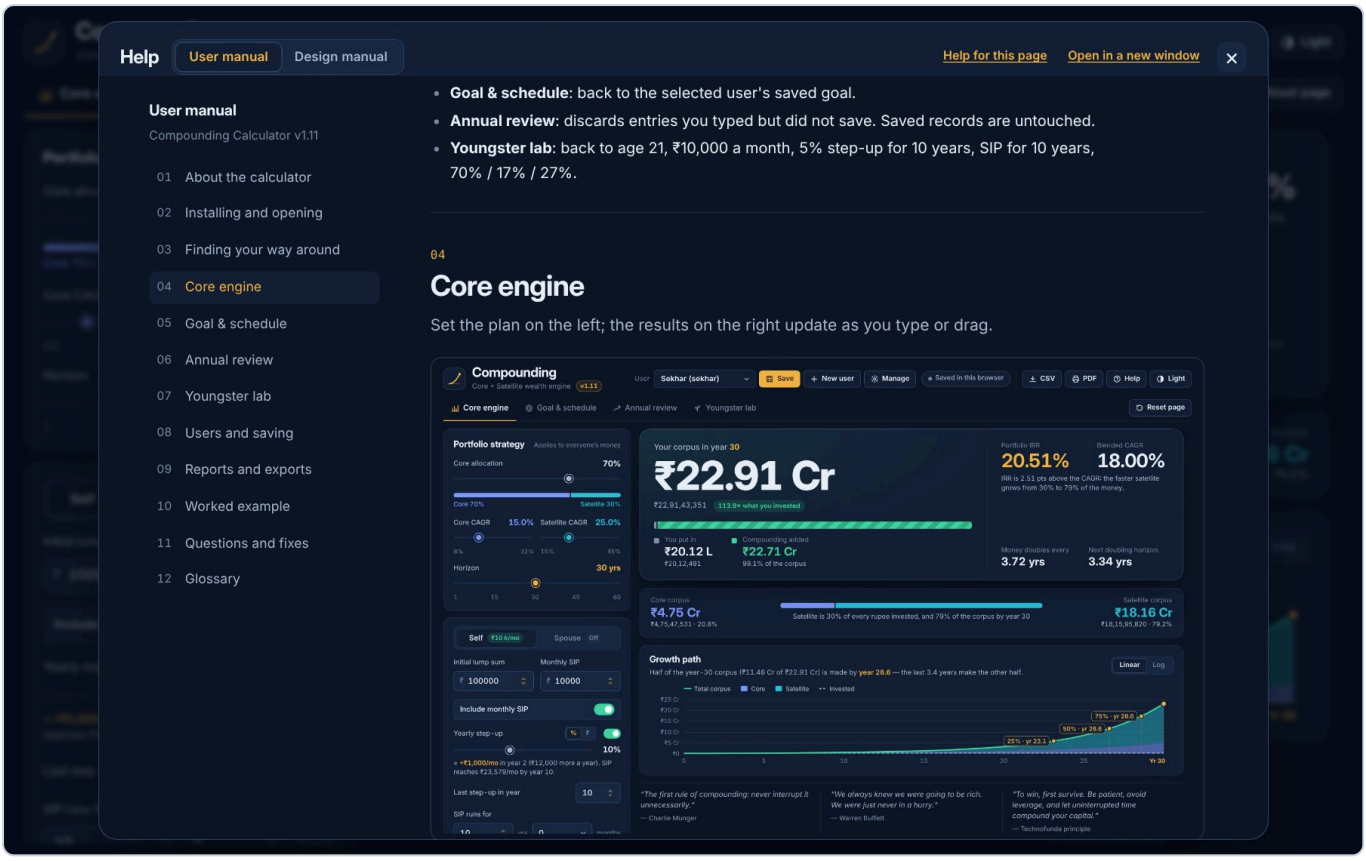
Open the link you were given. Nothing is installed. Whatever you save stays in that browser on that device, and nobody else can see it.

Your first five minutes

1. On **Core engine**, enter your lump sum and monthly SIP, then move the **Core allocation**, **Core CAGR** and **Satellite CAGR** sliders.
2. Read the large figure: your corpus in the final year. Hover over the chart to see any year.
3. Press **New user**, enter your name, a user ID and a goal, then **Create user**.
4. Open **Goal & schedule** to see when you reach the goal and the SIP that gets there on time.
5. Press **Save** whenever you want the current plan kept for this user.

HELP IS BUILT IN

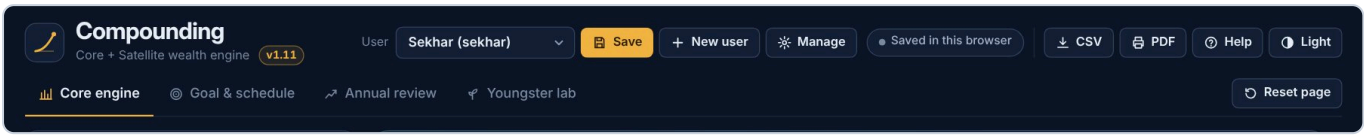
Press **Help** in the header (or `F1`) on any page. This manual opens at the chapter for that page. The Design manual is one click away in the same window.



Help window. Switch manuals at the top; **Help for this page** jumps back to the current page's chapter; **Open in a new window** shows the manual on its own.

03

Finding your way around



Header and page tabs. The header is the same on every page.

Control	What it does
Version pill (v1.11)	Shows the version, release date, developer and version history.
User	Switches between saved users. Each user has a plan, a goal and year-end records.
Save	Saves the current plan and goal to the selected user. It turns gold when there are unsaved changes.
New user	Creates a user with a name, user ID and goal.
Manage	Lists users; opens, deletes, exports and imports them; links a database file.
Storage pill	Says where data is saved: *Saved in this browser*, the database file name, or *Reconnect ...*. Shows • *unsaved* when there are changes not yet saved.
CSV	Downloads the 60-year schedule as a spreadsheet file (Windows app).
PDF	Prints the plan as a one-page summary, or a summary with the full schedule (Windows app).
Help	Opens this manual at the page you are on. `F1` does the same.
Light / Dark	Switches the colour theme. The choice is remembered.

Pages

The four tabs under the header are **Core engine**, **Goal & schedule**, **Annual review** and **Youngster lab**. All pages work from the same plan, so a change on Core engine shows up on the other pages at once.

Reset page

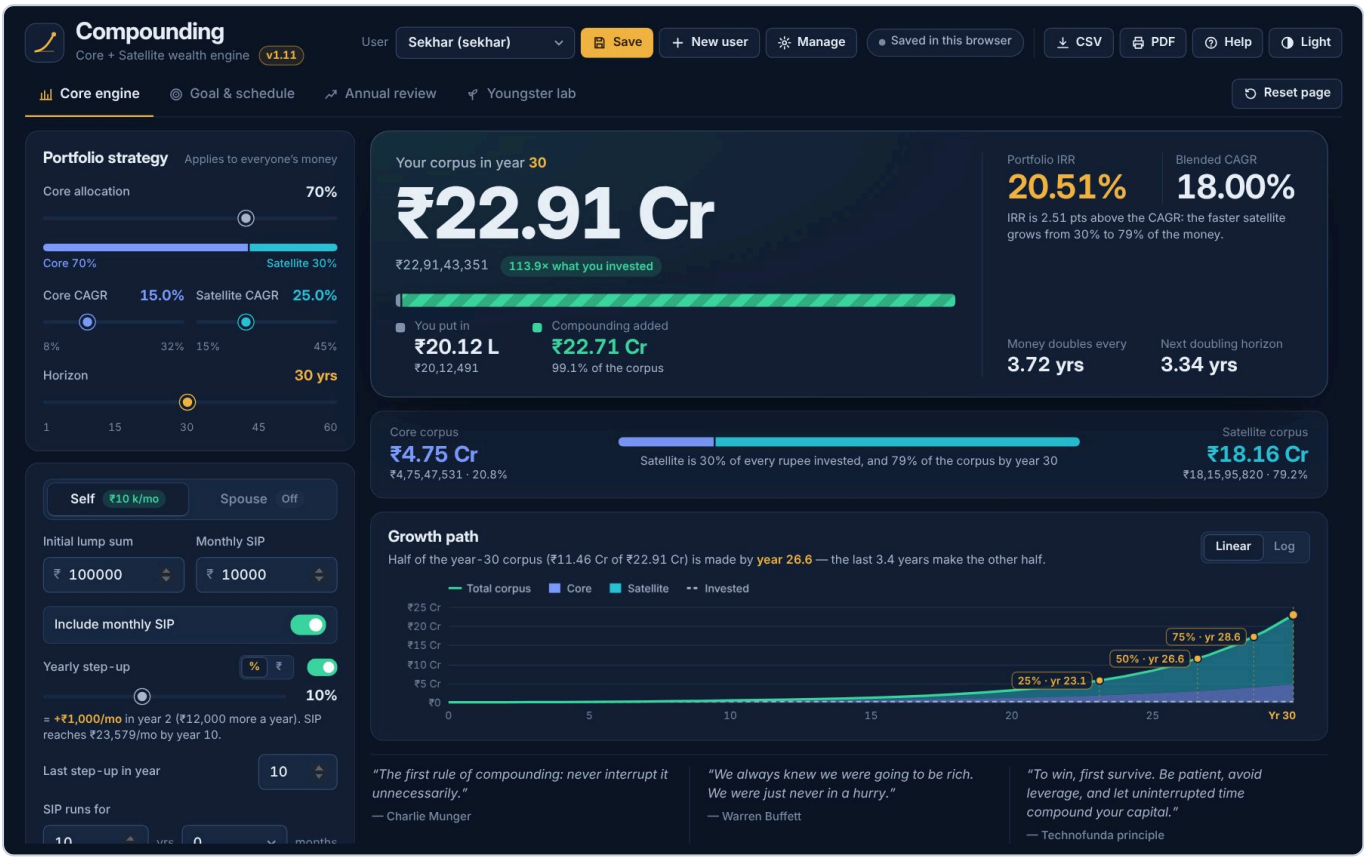
Reset page (right of the tabs) affects only the page you are on, and a message with **Undo** appears for a few seconds:

- **Core engine**: back to the standard plan (₹1,00,000 lump sum, ₹10,000 SIP, 10% step-up, 70/30, 15% / 25%, 30 years, spouse off).
- **Goal & schedule**: back to the selected user's saved goal.
- **Annual review**: discards entries you typed but did not save. Saved records are untouched.
- **Youngster lab**: back to age 21, ₹10,000 a month, 5% step-up for 10 years, SIP for 10 years, 70% / 17% / 27%.

04

Core engine

Set the plan on the left; the results on the right update as you type or drag.



Core engine with the standard plan. ₹1 lakh lump sum and ₹10,000 a month grow to about ₹22.91 Cr in 30 years.

Portfolio strategy

These settings apply to everyone's money in the plan, yours and your spouse's.

Control	Meaning	Range	Default
Core allocation	Share of every rupee that goes to the core. The rest goes to the satellite. The bar under it shows the split.	0–100%, steps of 5	70%
Core CAGR	Yearly return expected from the core.	8–32%	15%
Satellite CAGR	Yearly return expected from the satellite.	15–45%	25%
Horizon	The year the headline results are measured at.	1–60 yrs	30 yrs

Your money and your spouse's

The **Self** and **Spouse** buttons switch the input panel. Each shows a chip with the monthly amount, or *Off*.

Control	Meaning	Default
Initial lump sum	Money invested at the start (year 0).	₹1,00,000
Monthly SIP	Amount invested every month in year 1.	₹10,000
Include monthly SIP	Switch off to see the lump sum on its own.	On
Yearly step-up	How much the monthly SIP rises each year. Choose **%** (e.g. 10% more each year) or **₹** (e.g. ₹1,000 more each year). The line under it shows the other form, e.g. "= + ₹1,000/mo in year 2 … SIP reaches ₹23,579/mo by year 10". The switch turns step-up on or off.	On, 10%
Last step-up in year	The SIP stops rising after this year and stays flat.	10
SIP runs for	Years of SIP, plus 0–11 extra months. After this the money keeps compounding with no new money.	10 yrs 0 months
Add spouse's investments	Spouse panel only. Turns the spouse's lump sum and SIP on, with their own step-up and stop settings.	Off

TWO DIFFERENT STOPS

Last step-up in year ends the yearly rise; the SIP carries on at its last amount. **SIP runs for** ends the SIP itself. With both at 10, the SIP rises for 10 years and then stops.

Results

Your corpus in year *n*

Total value at the horizon, in crore and in full rupees, with how many times what you invested.

You put in

Everything invested: lump sums plus all SIPs up to the horizon.

Compounding added

Corpus minus what you put in, and its share of the corpus.

Portfolio IRR

The single yearly return that turns your actual cash flows (lump sum, then each year's SIPs) into the final corpus. This is the return your money really earned.

Blended CAGR

The planned average return: core share × core CAGR + satellite share × satellite CAGR. With 70/30, 15% and 25% it is 18%.

Money doubles every

Doubling time at the Portfolio IRR.

Next doubling horizon

Years after the horizon for the corpus to double once more.

Core and satellite corpus

Value of each pot at the horizon and its share of the wealth. The sentence between them compares the starting split with the ending split.

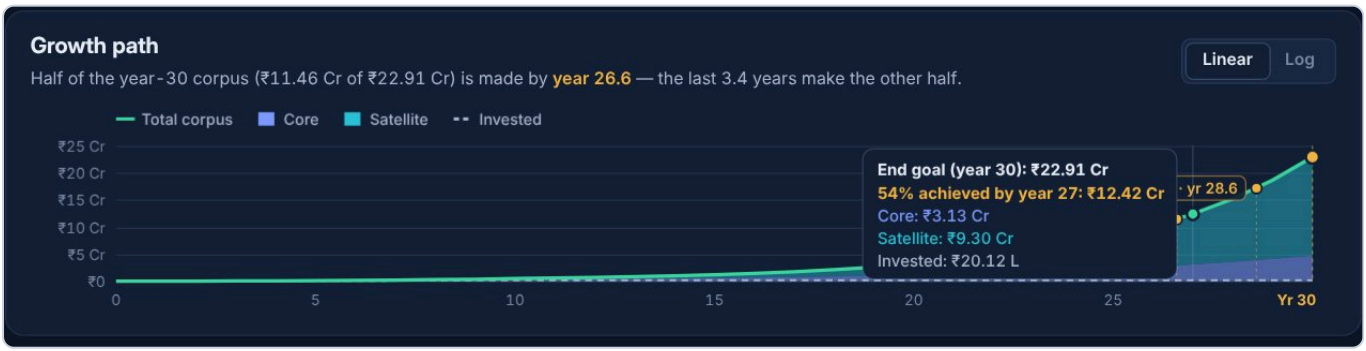
WHY IRR IS HIGHER THAN BLENDED CAGR

The pots are never rebalanced, so the faster satellite becomes a larger share of the money every year (30% of each rupee invested, but about 79% of the corpus by year 30). The return on the whole portfolio drifts up towards the satellite's rate. The note under the two figures states the gap in points.

Growth path chart

Lines show the total corpus, the core, the satellite and the money invested (dashed). **Linear** shows rupees to scale. **Log** makes each doubling the same height, which makes the early years readable.

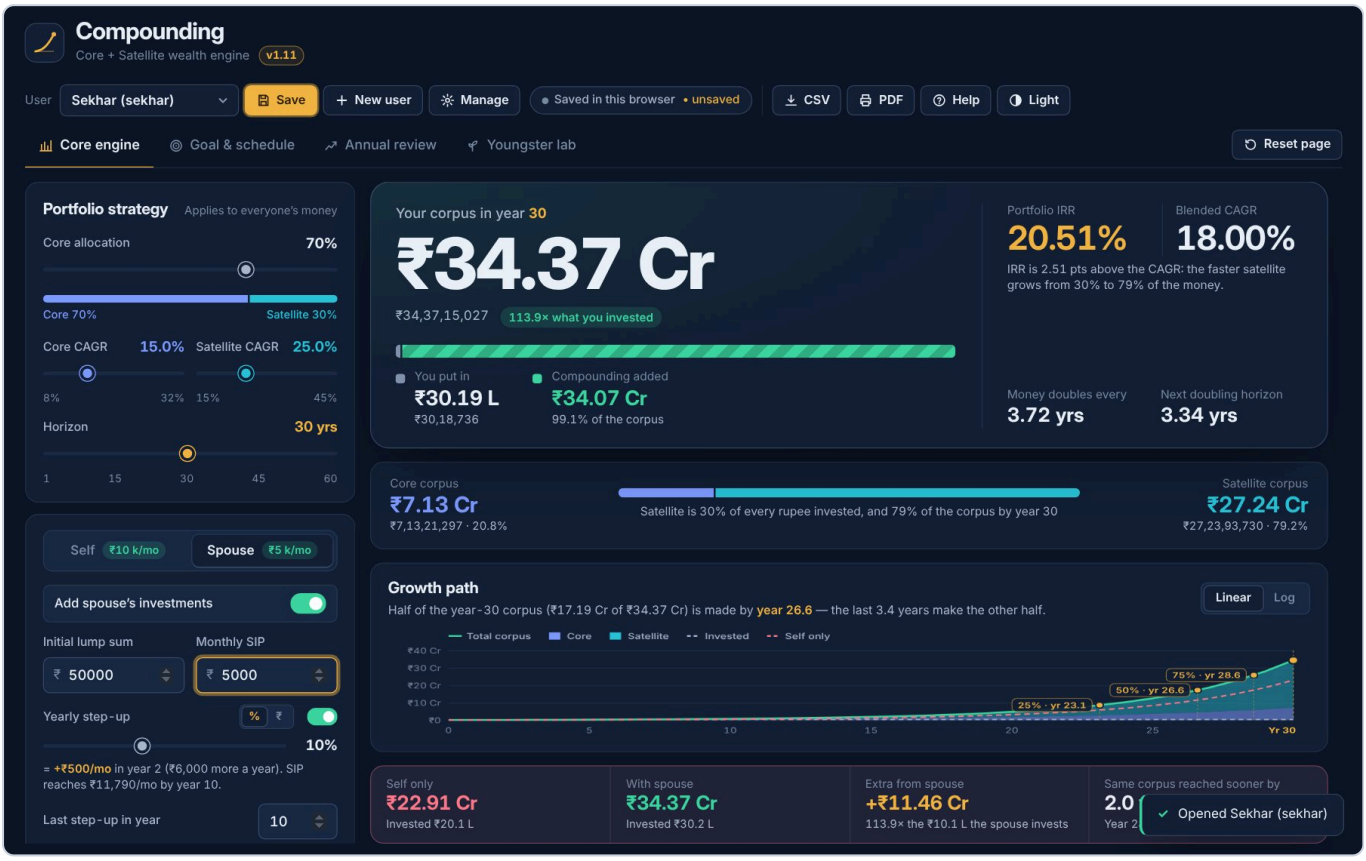
The gold markers show when **25%, 50% and 75%** of the final corpus has been made. The line above the chart says it in words: with the standard plan, half of the year-30 corpus is made by year 26.6, so the last 3.4 years make the other half.



Hover over the chart to read any year: End goal (year *n*), % achieved by that year, Core, Satellite and Invested.

With your spouse

When the spouse switch is on, a strip under the chart compares **Self only** with **With spouse**: both corpora, both amounts invested, the extra wealth from the spouse's money, and how much sooner the self-only corpus is reached together. The tooltip adds a *Self only* line.

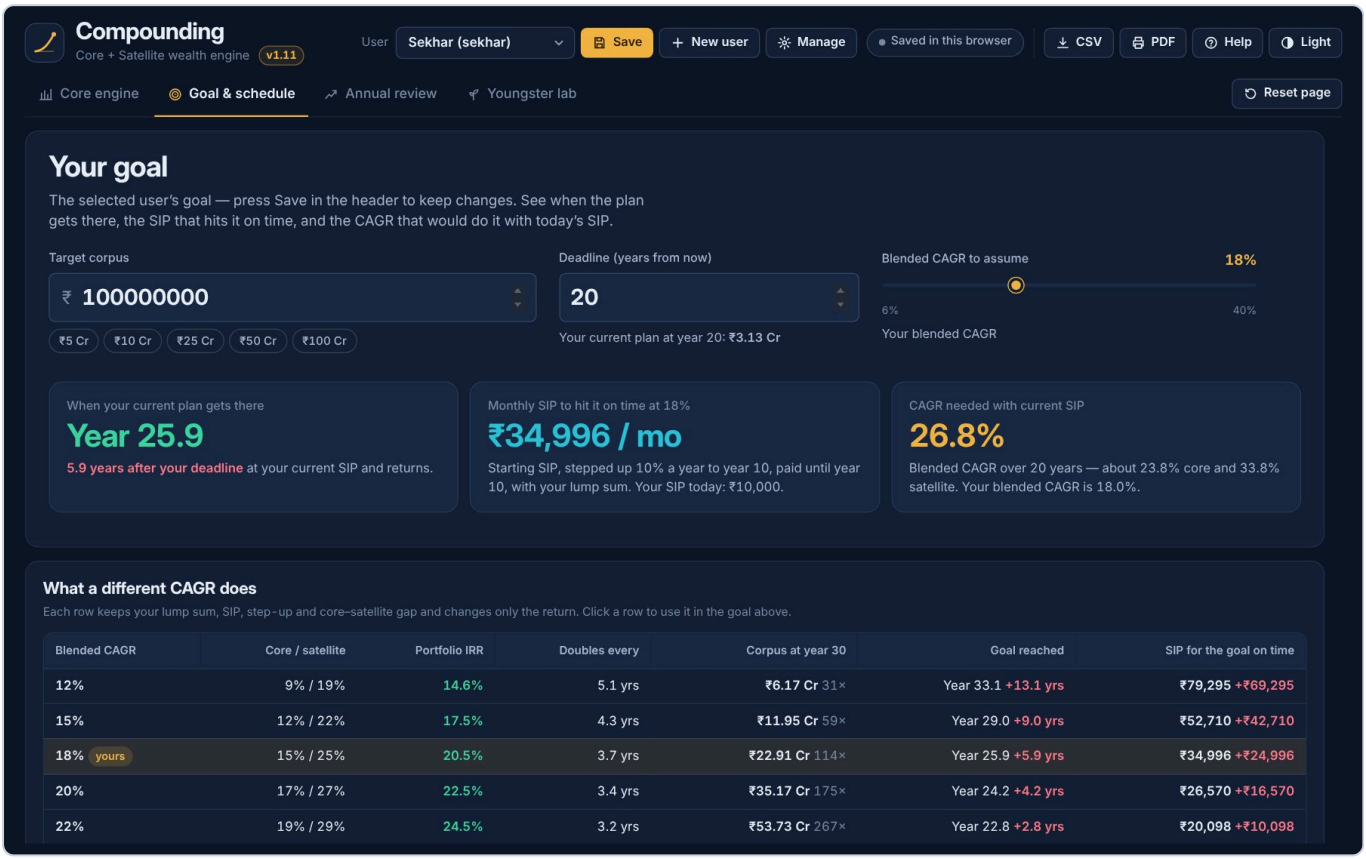


Spouse on: ₹50,000 lump sum and ₹5,000 a month added.

05

Goal & schedule

The selected user's goal, three ways to reach it, and every year of the plan.



Goal of ₹10 Cr in 20 years with the standard plan.

Setting the goal

Target corpus

The amount you want. The chips fill in ₹5, 10, 25, 50 or 100 Cr.

Deadline

Years from the start of the plan. The line below shows what the current plan has by then.

Blended CAGR to assume

The return used to work out the SIP needed (6–40%). It starts at your blended CAGR. Move it to test a more cautious or a bolder return; **Back to my CAGR** undoes it and **Use ... in my plan** copies it into the Core engine, keeping your core–satellite gap.

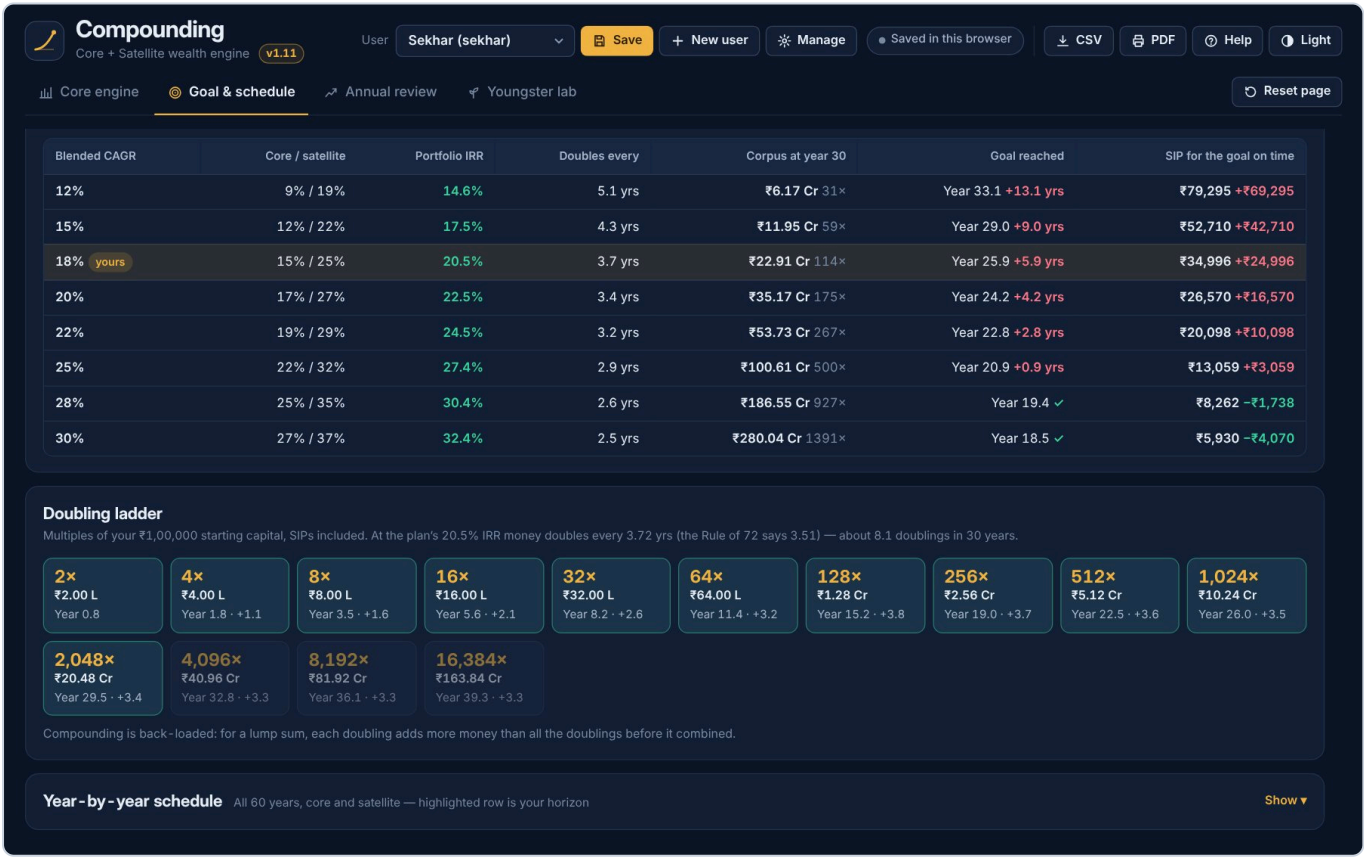
Goal changes belong to the selected user. Press **Save** in the header to keep them.

The three answers

Card	What it tells you	Standard plan, ₹10 Cr in 20 yrs
When your current plan gets there	The year the corpus first reaches the target, and how far that is from the deadline.	Year 25.9, 5.9 years late
Monthly SIP to hit it on time	The starting SIP, with your step-up, stop years and lump sum unchanged, that reaches the target exactly at the deadline at the assumed CAGR.	₹34,996 a month
CAGR needed with current SIP	The blended return that reaches the target on time with today's SIP, and the core and satellite rates that make it up.	26.8% (about 23.8% core, 33.8% satellite)

What a different CAGR does

Each row keeps your lump sum, SIP, step-up and core–satellite gap and changes only the return (12% to 30%, plus your own). Columns: core / satellite rates, Portfolio IRR, doubling time, corpus at your horizon, the year the goal is reached and the SIP needed for the goal on time. Your row is marked `yours`. Click any row to use that CAGR in the goal above.



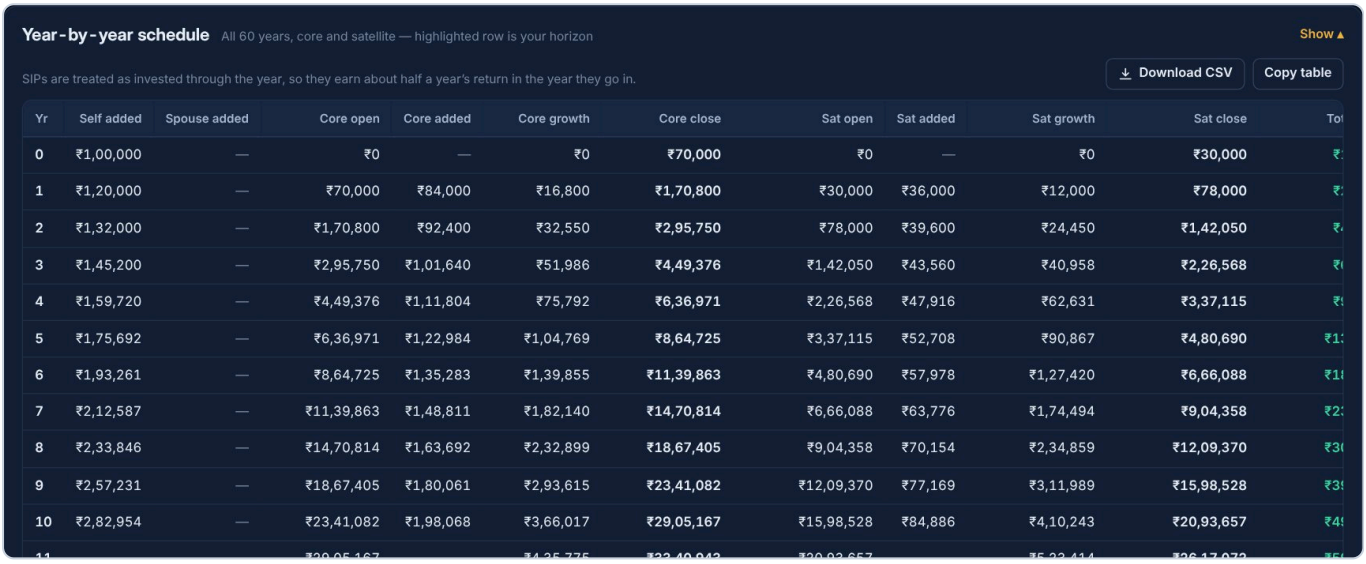
CAGR comparison and doubling ladder.

Doubling ladder

Each rung is a multiple of your starting capital (2×, 4×, 8× …) with the amount, the year it is reached and the gap from the previous rung. Rungs within your horizon are highlighted. The note above states the IRR, the doubling time and the Rule of 72 estimate for comparison.

Year-by-year schedule

Click **Show** to open the full 60-year table. Your horizon row is highlighted.



Yr	Self added	Spouse added	Core open	Core added	Core growth	Core close	Sat open	Sat added	Sat growth	Sat close	Tot
0	₹1,00,000	—	₹0	—	₹0	**₹70,000**	₹0	—	₹0	**₹30,000**	₹
1	₹1,20,000	—	₹70,000	₹84,000	₹16,800	**₹1,70,800**	₹30,000	₹36,000	₹12,000	**₹78,000**	₹
2	₹1,32,000	—	₹1,70,800	₹92,400	₹32,550	**₹2,95,750**	₹78,000	₹39,600	₹24,450	**₹1,42,050**	₹
3	₹1,45,200	—	₹2,95,750	₹1,01,640	₹51,986	**₹4,49,376**	₹1,42,050	₹43,560	₹40,958	**₹2,26,568**	₹
4	₹1,59,720	—	₹4,49,376	₹1,11,804	₹75,792	**₹6,36,971**	₹2,26,568	₹47,916	₹62,631	**₹3,37,115**	₹
5	₹1,75,692	—	₹6,36,971	₹1,22,984	₹1,04,769	**₹8,64,725**	₹3,37,115	₹52,708	₹90,867	**₹4,80,690**	₹1
6	₹1,93,261	—	₹8,64,725	₹1,35,283	₹1,39,855	**₹11,39,863**	₹4,80,690	₹57,978	₹1,27,420	**₹6,66,088**	₹1
7	₹2,12,587	—	₹11,39,863	₹1,48,811	₹1,82,140	**₹14,70,814**	₹6,66,088	₹63,776	₹1,74,494	**₹9,04,358**	₹2
8	₹2,33,846	—	₹14,70,814	₹1,63,692	₹2,32,899	**₹18,67,405**	₹9,04,358	₹70,154	₹2,34,859	**₹12,09,370**	₹3
9	₹2,57,231	—	₹18,67,405	₹1,80,061	₹2,93,615	**₹23,41,082**	₹12,09,370	₹77,169	₹3,11,989	**₹15,98,528**	₹3
10	₹2,82,954	—	₹23,41,082	₹1,98,068	₹3,66,017	**₹29,05,167**	₹15,98,528	₹84,886	₹4,10,243	**₹20,93,657**	₹4

The schedule, with **Download CSV** and **Copy table**.

Column	Meaning
Self added / Spouse added	SIP money put in that year.
Core open, added, growth, close	The core pot at the start of the year, money added, growth earned, and value at year end. The same four for the satellite.
Total corpus / Total invested	Both pots together, and everything put in so far.
Multiple	Corpus ÷ invested.
% of final corpus	How much of the horizon corpus exists by that year.
Yrs to next 2×	Years until the corpus doubles from that year's value.

SIPs are treated as invested through the year, so in the year they go in they earn about half a year's return.

06

Annual review

Once a year, record what you actually invested and what the portfolio is worth. The calculator re-plans the rest of the road to the goal.



Annual review for an example user with two years recorded.

Choose the user

Type a name or user ID in **Find a user by name or user ID** and press **Open** (or pick from the header). The user card shows the goal, the year the plan started, how many years are recorded and when the user was last saved. **Edit user & goal** changes the name, goal or start year.

The four cards

Projected at goal year

Recorded years as they actually happened, then the plan for the rest. Says how far above or short of the goal this lands.

Goal reached in

The year the revised projection reaches the goal, before or after the deadline.

Return needed from here on

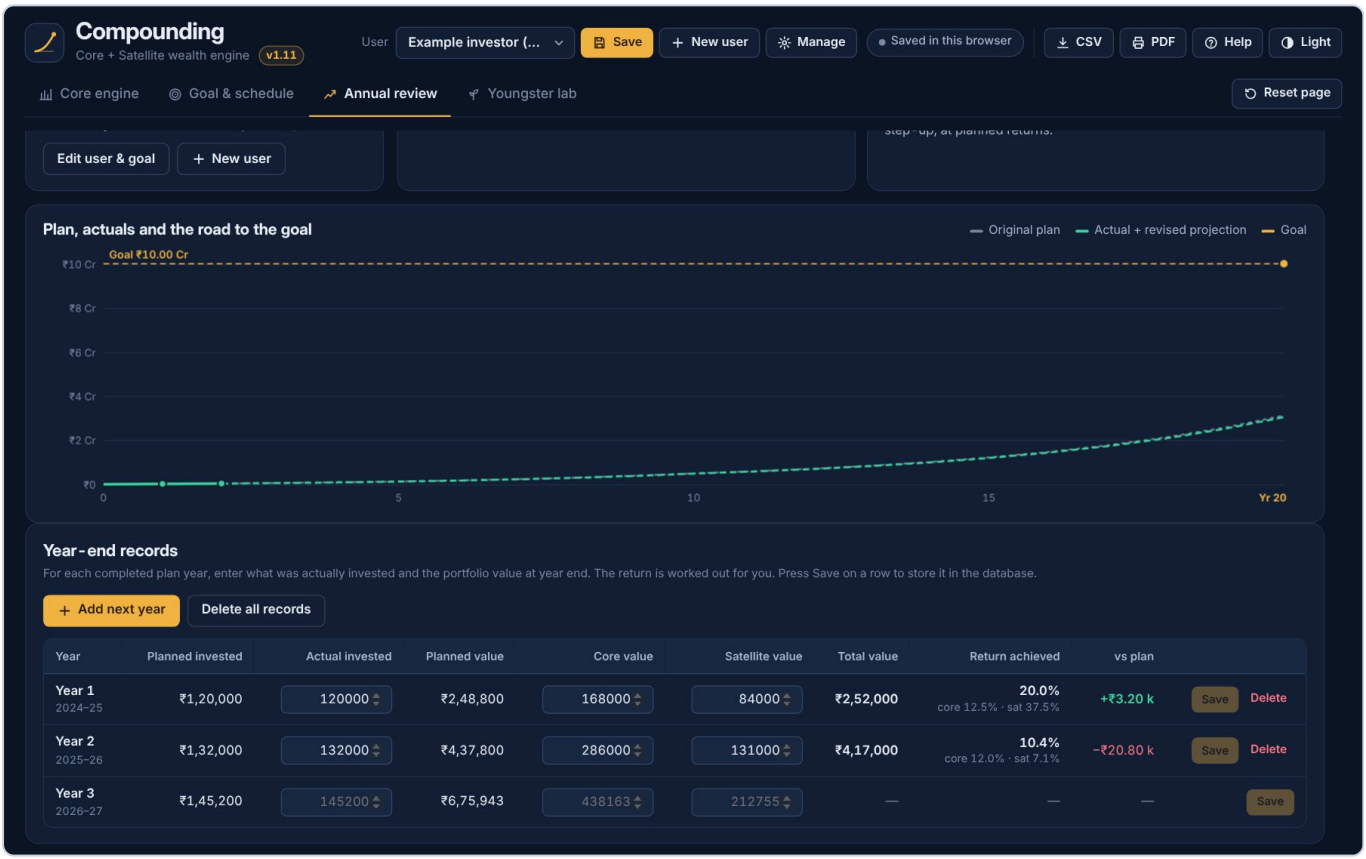
The blended CAGR needed for the remaining years, compared with your planned CAGR.

Monthly SIP needed from year *n*

The SIP from next year on that reaches the goal at the planned returns. If your SIP is still running, the planned SIP path (with its step-up) is scaled up or down. If it has already ended, the card shows a flat SIP restarted until the goal year. Past years are never changed.

Recording a year

1. At the end of each plan year, find that year's row in **Year-end records**. The grey placeholders show the planned figures.
2. Enter **Actual invested** (leave blank to use the planned amount), **Core value** and **Satellite value** at year end. Both values are needed.
3. Press **Save** on the row (or `Enter`). The record is stored for the user straight away, and the next year's row appears.



Year-end records. Return achieved is worked out for you; *vs plan* shows the difference from the original plan.

Return achieved is the year's growth divided by the money at work: opening value plus half of the year's investments. It is shown for the whole portfolio and for core and satellite separately. A row's **Delete** removes that year; **Delete all records** clears the user's history.

RECORDS ALSO MOVE THE CORE ENGINE

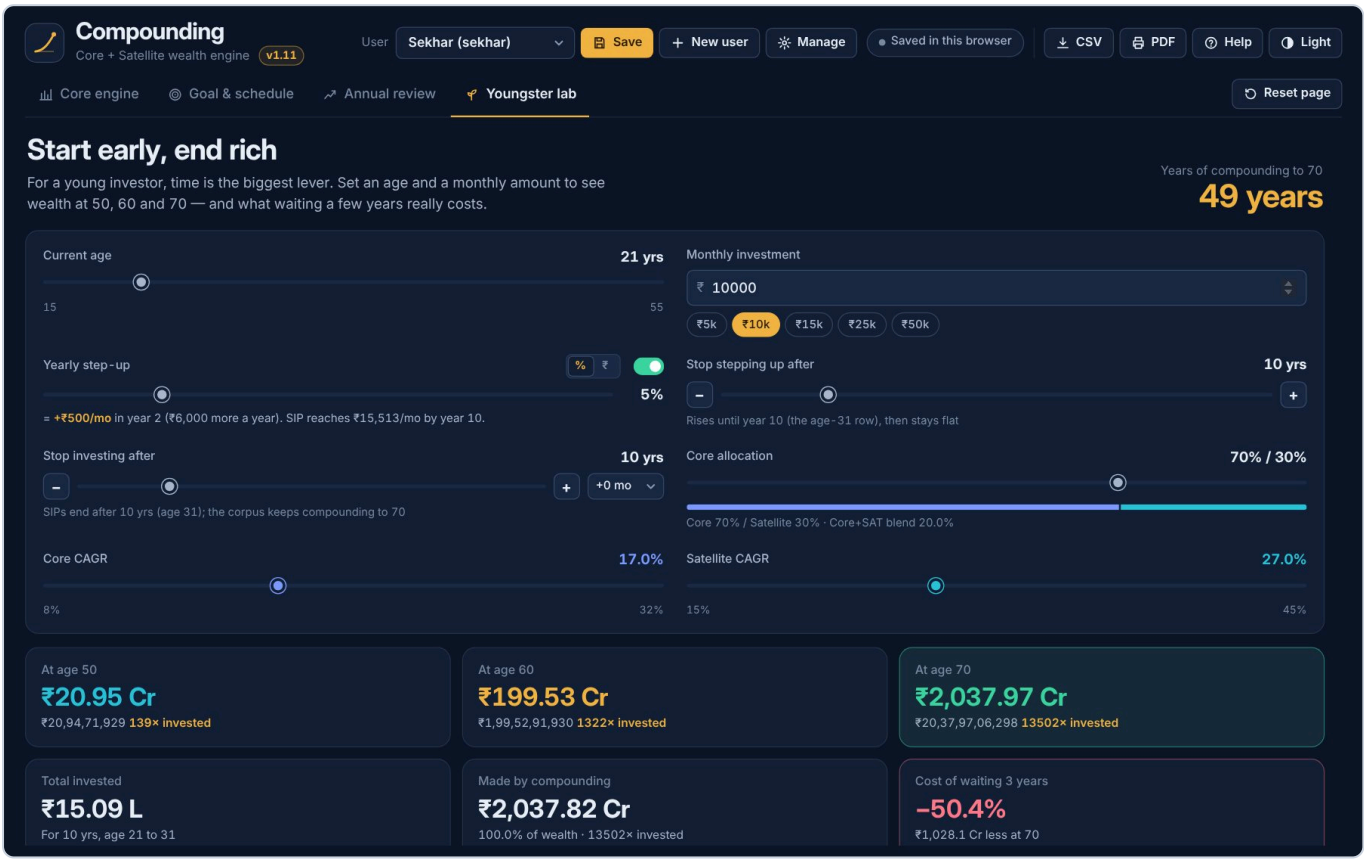
Once a user has records, the Core engine and Goal & schedule pages start from the recorded values too, so all pages tell the same story.

The chart compares the **original plan**, the **actual + revised projection** and the **goal** line.

07

Youngster lab

For a young investor: wealth at 50, 60 and 70, and what waiting three, six or nine years really costs.



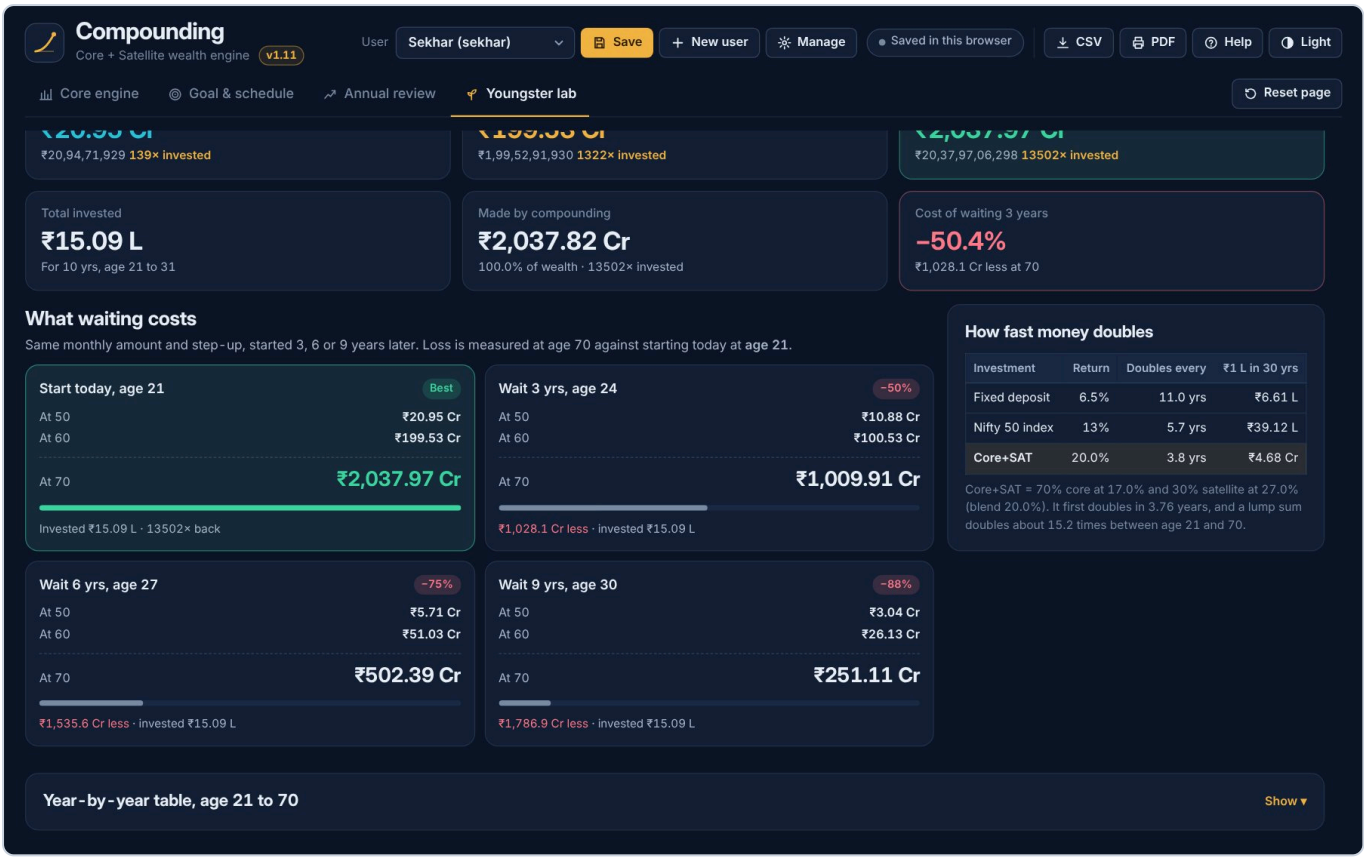
Youngster lab with its starting values.

Inputs

Control	Meaning	Default
Current age	15 to 55. The years of compounding to 70 show top right.	21
Monthly investment	Type an amount or use the chips (₹5k to ₹50k).	₹10,000
Yearly step-up	In % or ₹, with an on/off switch, as on the Core engine.	5%
Stop stepping up after	Use – and + for 1-year steps. After this the SIP stays flat.	10 yrs
Stop investing after	Years plus 0–11 months. The corpus keeps compounding to 70.	10 yrs
Core allocation	Core / satellite split, with the Core+SAT blend shown under the bar.	70%
Core CAGR / Satellite CAGR	Expected return of each pot.	17% / 27%

Results

Cards show wealth at **50**, **60** and **70** with the multiple of money invested, the total invested, the amount made by compounding, and the **cost of waiting 3 years**. With the defaults: ₹20.95 Cr at 50, ₹199.53 Cr at 60 and ₹2,037.97 Cr at 70 from ₹15.09 L invested.



What waiting costs and **How fast money doubles**.

What waiting costs

The same plan started 3, 6 or 9 years later, measured at age 70. With the defaults, waiting 3 years costs about half the final wealth (−50%), 6 years −75% and 9 years −88%.

How fast money doubles

A small table compares a fixed deposit (6.5%), the Nifty 50 index (13%) and your **Core+SAT** mix from the three sliders: the return, the doubling time and what ₹1 lakh becomes in 30 years. The note under it gives the mix's first doubling time and how many times a lump sum doubles before 70.

> **WHY THE AGE-70 FIGURE IS SO LARGE**
>
> The core and the satellite grow as separate pots with no rebalancing, as in the Core engine. Over 49 years the 27% satellite grows far faster and ends up as nearly all the money, so the result is much higher than a flat 20% would give (₹491.80 Cr). Set Core allocation to 100% and Core CAGR to 20% to see the flat-rate figure.

Year-by-year table

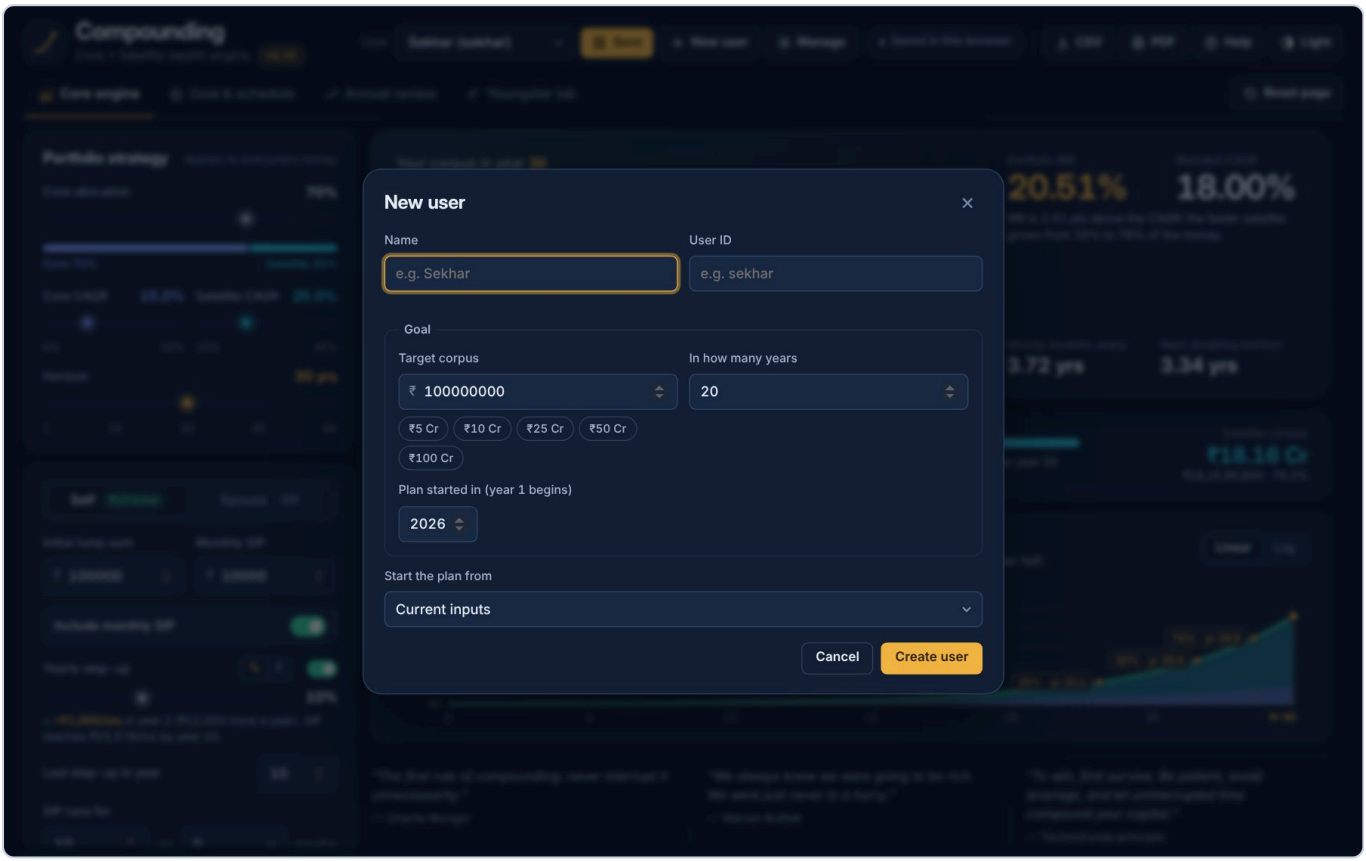
Click **Show** to see every year from your age to 70: age, year, monthly SIP, invested and growth that year, total invested, wealth and multiple.

08

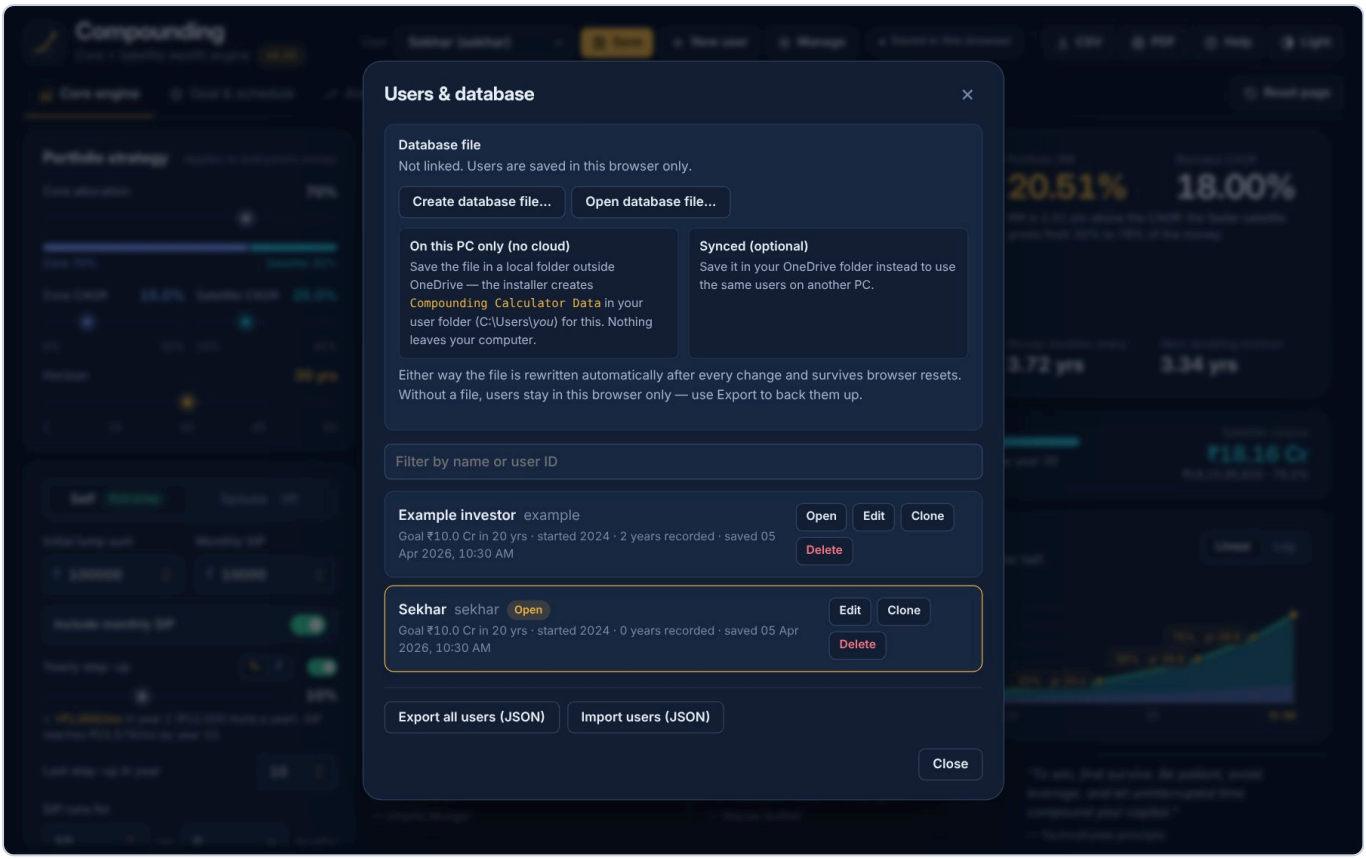
Users and saving

Each user holds one plan, one goal and their year-end records. Users work like a small database kept on your computer.

Creating a user



New user.



Users & database (Manage).

Name

Shown in the user list and on reports.

User ID

A unique short ID: 2–32 characters of lowercase letters, digits, dot, dash or underscore (e.g. `sekhar` or `priya.k`). It is suggested from the name.

Goal

Target corpus, the number of years, and the year the plan started (year 1).

Start the plan from

Current inputs copies the Core engine as it is now; *Standard plan* uses ₹1 L + ₹10k SIP, 15% / 25%.

Saving

Plan and goal changes are kept for a user only when you press **Save**. Until then the button is gold and the storage pill says • *unsaved*. Year-end records are different: each row is saved when you press its own **Save**. Switching user with unsaved changes asks first.

Where data is kept

Users are always saved in the browser the calculator runs in. In the Windows app you can also link a **database file** (Manage → Database file), which is rewritten after every change and survives browser resets:

- **On this PC only**: create the file in `Compounding Calculator Data` in your user folder. Nothing leaves your computer.
- **Synced**: create it in your OneDrive folder to use the same users on another PC. Open the same file there with **Open database file...**.

When two copies differ, the more recently updated version of each user wins, and deleted users stay deleted. After restarting Windows the pill may say *Reconnect <file>*; click it and allow access.

Backups

Export all users (JSON) saves every user to one file; **Import users (JSON)** merges a file back in using the same newer-wins rule. Export before reinstalling Windows or moving to a new PC if you do not use a database file.

09

Reports and exports

PDF report

1. Press **PDF** in the header.
2. Choose **Summary (1 page)**: assumptions, headline numbers, growth chart and doubling milestones; or **Summary + 60-year schedule**, which adds the full table with headers repeated on every page.
3. Press **Open print dialog**, choose **Save as PDF** and turn on **Background graphics** for colours.

The report header shows the investor (the selected user), the date, the horizon, the version and the developer.

CSV and copying

CSV in the header, or **Download CSV** in the schedule, saves the 60-year schedule for Excel. **Copy table** copies it so you can paste it into a spreadsheet or email.

The web edition cannot download files or open the print dialog, so these buttons appear only in the Windows app.

10

Worked example: the standard plan

These are the figures you see after **Reset page** on Core engine, with no year-end records.

Input	Value
Lump sum / monthly SIP	₹1,00,000 / ₹10,000
Step-up	10% a year, last in year 10
SIP runs for	10 years
Split and returns	70% core at 15%, 30% satellite at 25%
Horizon	30 years

How the SIP grows

Year 1: ₹10,000 a month (₹1,20,000 for the year). Year 2: ₹11,000. Year 3: ₹12,100. By year 10: ₹23,579 a month. From year 11 nothing new is added. Total put in: **₹20,12,491**.

What it grows to

Year	Invested so far	Core	Satellite	Corpus
1	₹2.20 L	₹1.71 L	₹0.78 L	₹2.49 L
10	₹20.12 L	₹29.05 L	₹20.94 L	₹49.99 L
20	₹20.12 L	₹1.18 Cr	₹1.95 Cr	₹3.13 Cr
30	₹20.12 L	₹4.75 Cr	₹18.16 Cr	**₹22.91 Cr**

- **Portfolio IRR 20.51%** against a **Blended CAGR of 18.00%**: the satellite grows from 30% of the money to 79% of the corpus.
- Money doubles every **3.72 years**; the next doubling after year 30 takes **3.34 years**.
- 25% of the year-30 corpus is made by year 23.1, 50% by year 26.6 and 75% by year 28.6.
- Compounding adds ₹22.71 Cr, about 99% of the corpus, on ₹20.12 L invested.

Against a goal of ₹10 Cr in 20 years

The plan has ₹3.13 Cr at year 20 and reaches ₹10 Cr in year 25.9. To get there on time you would need either a starting SIP of **₹34,996** a month (same step-up and stop years), or a blended CAGR of **26.8%** with today's SIP.

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Questions and fixes

My corpus changed after I entered a year-end record.

That is intended. Recorded years replace the plan for that user on every page. Open a user with no records, or delete the records, to see the pure plan.

Why is the Portfolio IRR different from the Blended CAGR?

Because the satellite is not rebalanced back to 30%, it becomes a growing share of the portfolio. See chapter 4.

The Goal card says “Not reachable”.

The SIP stops before the deadline, so no starting SIP gets there in time. Increase **SIP runs for** on the Core engine, extend the deadline, or lower the target.

The storage pill says “Reconnect …”.

Windows needs your permission again to edit the database file. Click the pill and choose **Allow**. Until then changes are kept in the browser and written to the file once reconnected.

My users are missing.

Users live in the browser that saved them. If you opened the calculator in a different browser or profile, use the database file or **Import users (JSON)** from a backup.

In the web edition, Delete does nothing.

Pop-up confirmations are blocked there. Click the same Delete button again within 6 seconds to confirm.

The months option is on SIP but not on step-up.

By design: the SIP can end part-way through a year, while the step-up changes only once a year.

Can returns be negative?

Recorded years can show negative returns. Planned returns are always within the slider ranges.

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Glossary

Lakh, crore
₹1 lakh (L) = ₹1,00,000. ₹1 crore (Cr) = ₹1,00,00,000 = 100 lakh. Amounts are grouped the Indian way.

Corpus
The total value of the portfolio at a point in time.

Lump sum
Money invested once, at the start.

SIP
Systematic investment plan: a fixed amount invested every month.

Step-up
The yearly increase in the monthly SIP, in % or ₹.

Core
The larger, steadier part of the portfolio.

Satellite
The smaller, higher-growth part of the portfolio.

CAGR
Compound annual growth rate: the steady yearly return assumed for a pot.

Blended CAGR
The allocation-weighted average of core and satellite CAGR.

IRR
Internal rate of return: the one yearly return that links every rupee put in, at the time it went in, to the final corpus.

Horizon
The year results are measured at.

Multiple
Corpus divided by the total amount invested.

Doubling time
Years for money to double at a given return: ln 2 ÷ ln(1 + return).

Rule of 72
A quick estimate of doubling time: 72 ÷ return in %.

Return achieved
In the Annual review: a year's growth divided by the opening value plus half the year's investments.